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import torch
import torch.nn as nn
import torch.optim as optim
import torchvision
import torchvision.transforms as transforms

# Load dataset
transform = transforms.ToTensor()
trainset = torchvision.datasets.MNIST(root='./data', train=True, download=True, transform=transform)
trainloader = torch.utils.data.DataLoader(trainset, batch_size=64, shuffle=True)
testset = torchvision.datasets.MNIST(root='./data', train=False, download=True, transform=transform)
testloader = torch.utils.data.DataLoader(testset, batch_size=64, shuffle=False)

# Define Feed Forward Neural Network
class FFNN(nn.Module):
    def __init__(self):
        super(FFNN, self).__init__()
        self.fc1 = nn.Linear(28*28, 128)
        self.fc2 = nn.Linear(128, 64)
        self.fc3 = nn.Linear(64, 10)
        self.relu = nn.ReLU()
        self.softmax = nn.LogSoftmax(dim=1)

    def forward(self, x):
        x = x.view(-1, 28*28)
        x = self.relu(self.fc1(x))
        x = self.relu(self.fc2(x))
        x = self.fc3(x)
        return self.softmax(x)

model = FFNN()
criterion = nn.CrossEntropyLoss()
optimizer = optim.Adam(model.parameters(), lr=0.001)

# Training
for epoch in range(5):
    running_loss = 0.0
    correct = 0
    total = 0
    for images, labels in trainloader:
        optimizer.zero_grad()
        outputs = model(images)
        loss = criterion(outputs, labels)
        loss.backward()
        optimizer.step()
        running_loss += loss.item()

    _, predicted = torch.max(outputs.data, 1)
    total += labels.size(0)
    correct += (predicted == labels).sum().item()

acc = 100 * correct / total
print(f"Epoch {epoch+1}, Loss: {running_loss:.3f}, Accuracy: {acc:.2f}%")

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100%|██████████| 28.9k/28.9k [00:00<00:00, 339kB/s]
100%|██████████| 1.65M/1.65M [00:00<00:00, 3.18MB/s]
100%|██████████| 4.54k/4.54k [00:00<00:00, 8.27MB/s]
Epoch 1, Loss: 328.589, Accuracy: 90.29%
Epoch 2, Loss: 136.517, Accuracy: 95.67%
Epoch 3, Loss: 93.478, Accuracy: 97.06%
Epoch 4, Loss: 70.181, Accuracy: 97.73%
Epoch 5, Loss: 53.096, Accuracy: 98.22%

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